

Super Key

Definition

A Super Key is a set of one or more attributes (columns) that can uniquely identify every row in a table.

In simple words,

Any combination of columns that guarantees uniqueness is called a Super Key.

UNIQUENESS

NO DUPLICATE VALUES

CAN CONTAIN EXTRA ATTRIBUTES

CAN HAVE MANY SUPER KEYS

Candidate Key

A Candidate Key is a minimal Super Key.

In other words,

A Candidate Key is the smallest set of attributes that can uniquely identify every row in a table.

It has two important properties:

- It uniquely identifies every row.
- It contains no unnecessary attributes.

UNIQUENESS

MINIMALITY

NO REDUNDANT ATTRIBUTES

MULTIPLE

Why is minimality important?

Answer:

Minimality ensures that a key contains only the attributes required for uniqueness. Removing redundant attributes reduces storage overhead, simplifies relationships, and results in cleaner database design.

Primary Key

Definition

A Primary Key (PK) is the Candidate Key chosen to uniquely identify every row in a table.

Every table should have only one Primary Key.

Every Primary Key must be

- **UNIQUE**
- **NOT NULL**

Both are mandatory.

A Primary Key can never contain NULL because every row must have a valid and unique identifier. This is required by the Entity Integrity rule.

changing a Primary Key is strongly discouraged because it can require updating all related foreign keys and may affect indexes and application logic. Therefore, Primary Keys should be chosen to be stable.

A table may have multiple Candidate Keys, but having one designated Primary Key provides a single, consistent identifier for relationships, indexing, and database design. The remaining Candidate Keys become Alternate Keys.